

FLAGSTAFF PULLIAM, AZ, USA

WMO: 723750

Lat: 35.144N

Lon: 111.666W

Elev: 2135

StdP: 78.18

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03103

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-15.4	-12.3	-22.0	0.7	-3.7	-19.9	0.8	-3.3	12.6	0.6	11.4	1.1	0.6	40	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	29.9	12.9	28.7	12.7	27.4	12.6	16.4	22.5	15.7	22.3	15.2	22.0	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	13.6	17.7	13.9	12.9	17.3	13.1	12.2	17.0	53.8	22.3	51.7	22.2	49.9	21.8	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.8	8.5	-20.4	32.5	2.7	1.3	-22.4	33.4	-24.0	34.2	-25.5	34.9	-27.4	35.9		
(5)			-20.8	18.2	2.6	0.9	-22.7	18.8	-24.2	19.3	-25.7	19.8	-27.6	20.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-0.7	0.3	3.5	6.3	10.7	16.3	19.2	18.1	14.5	8.4	3.2	-1.0		
	DBStd	7.87	3.92	4.02	3.55	3.54	3.57	3.25	1.99	1.86	2.60	3.35	3.71	4.11		
	HDD10.0	1573	333	273	201	117	34	2	0	0	2	68	203	340		
	HDD18.3	3747	591	506	459	361	237	76	13	27	117	307	453	598		
	CDD10.0	944	0	0	0	6	57	189	286	251	137	19	0	0		
	CDD18.3	75	0	0	0	0	1	14	41	19	1	0	0	0		
	CDH23.3	1936	0	0	0	2	71	625	711	401	122	4	0	0		
	CDH26.7	431	0	0	0	0	8	153	189	71	9	0	0	0		
(14) Wind	WSAvg	3.0	2.8	3.1	3.3	3.8	3.6	3.4	2.4	2.0	2.4	2.7	2.8	2.9		
(15) Precipitation	PrecAvg	573	51	55	61	34	18	10	65	72	52	44	47	55		
	PrecMax	929	243	255	172	143	105	49	168	205	172	250	169	185		
	PrecMin	367	0	0	0	0	0	0	0	7	0	0	0	0		
	PrecStd	154	49	52	47	30	20	12	36	41	45	46	42	52		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	16.5	19.7	22.7	27.9	31.7	31.9	30.2	27.8	23.9	19.5	14.7		
		MCWB	4.2	5.2	6.9	8.2	11.1	12.6	12.9	13.7	12.7	9.9	7.0	4.4		
	2%	DB	11.6	13.9	17.2	20.5	25.3	29.7	30.2	28.6	26.1	21.8	17.1	12.0		
		MCWB	2.9	4.1	5.9	7.4	9.9	11.9	13.3	13.7	12.2	9.2	6.1	3.5		
	5%	DB	9.7	11.5	15.2	18.5	23.4	28.5	28.8	27.2	24.7	20.2	15.1	9.7		
		MCWB	2.1	3.0	4.9	6.4	9.1	11.5	13.2	13.8	11.9	8.6	5.5	2.2		
	10%	DB	7.3	9.0	13.3	16.7	21.6	27.1	27.4	25.7	23.3	18.4	12.8	7.2		
		MCWB	1.2	2.0	4.1	5.7	8.3	11.0	13.2	13.9	11.9	8.0	4.5	1.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	6.4	7.7	9.2	11.9	14.9	17.3	17.3	16.2	12.6	8.4	5.9		
		MCDB	10.4	12.4	17.0	20.4	24.1	24.3	22.8	22.9	21.9	17.7	16.1	11.9		
	2%	WB	3.9	5.0	6.4	8.0	10.7	13.6	16.5	16.4	15.1	11.1	7.0	4.5		
		MCDB	9.2	11.6	16.1	18.8	22.5	25.3	22.6	22.5	20.9	17.8	14.3	9.7		
	5%	WB	2.8	3.8	5.4	6.9	9.9	12.8	15.8	15.8	14.2	10.0	6.0	3.1		
		MCDB	7.8	10.1	14.0	17.2	21.4	25.5	22.5	22.2	20.0	17.5	13.4	7.7		
	10%	WB	1.6	2.6	4.4	6.0	8.9	11.9	15.3	15.3	13.4	8.8	5.1	1.8		
		MCDB	6.5	8.1	12.2	15.7	20.1	24.7	22.4	21.9	19.5	16.7	11.7	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.5	15.1	16.0	17.3	19.1	15.6	14.7	16.1	16.6	15.2	13.5		
		MCDBR	17.6	18.9	19.8	20.7	20.9	20.7	18.5	17.5	18.9	20.9	20.1	18.2		
		MCWBR	10.5	10.7	10.3	10.3	9.8	9.2	7.2	6.9	8.2	10.6	11.3	10.9		
		MCDBR	15.3	15.7	17.9	18.8	19.0	18.3	13.4	13.5	13.8	16.6	17.3	15.2		
	5% WB	MCWBR	9.5	9.4	9.7	9.7	9.0	8.1	5.3	5.9	6.6	8.9	10.1	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.232	0.235	0.254	0.264	0.278	0.280	0.326	0.328	0.283	0.268	0.238	0.232		
	taud		2.681	2.692	2.621	2.582	2.590	2.577	2.522	2.497	2.645	2.624	2.682	2.684		
	Ebn at Noon		1009	1036	1030	1024	1005	996	949	944	983	978	992	990		
	Edh at Noon		68	75	89	97	98	99	104	105	86	80	67	64		
(44) All-Sky Solar Radiation	RadAvg		3.17	4.12	5.62	7.02	7.85	8.36	6.75	6.24	5.87	4.70	3.59	2.82		
	RadStd		0.28	0.35	0.37	0.32	0.50	0.41	0.43	0.33	0.22	0.35	0.17	0.20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+98	+40

Nomenclature: See separate page